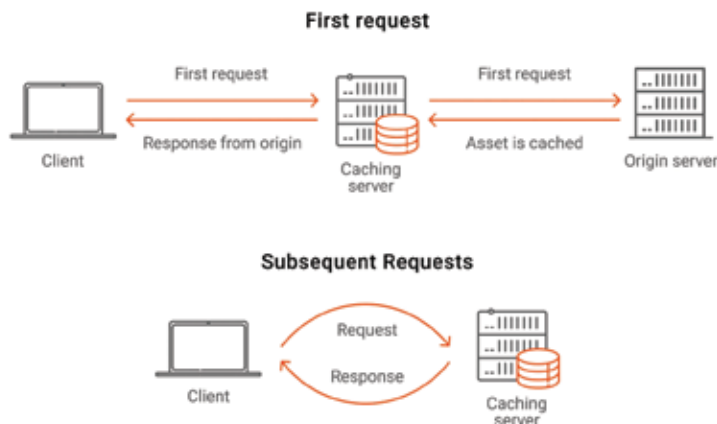




Much of the information you see when you log in, such as post data, friends, and so on, is cached by Facebook. Redis and Memcache are the two most widely used caching server technologies.

Job Queue & Servers

Most online applications require asynchronous work behind the scenes that isn't directly related to responding to a user's request. For example, in order to return search results, Google must crawl and index the entire internet.

This does not happen every time you conduct a search. Instead, it crawls the web in real-time, updating search indexes as it goes. While there are a variety of architectures that allow for asynchronous work, the most common is what we'll refer to as the "task queue" architecture.

It is made up of two parts: a queue of "jobs" that must be completed and one or more job servers (also known as "workers") who execute the jobs in the queue. A list of jobs that must be run asynchronously is stored in a job queue. First-in-first-out (FIFO) queues are the simplest, but most applications require some type of priority queueing scheme.

The app simply adds the necessary job to the queue whenever it needs to run a job, whether on a regular schedule or in response to user activities. For example, a website for an e-commerce portal could use a job queue to power a lot of the behind-the-scenes labour that keeps the marketplaces running.

It can execute jobs to process orders and initiate supply-chain logistics, process taxes for orders and issue customer rewards, aggregate user statistics and send notification emails, among other things.